

Necessary in Its Native Block: Selectivity and Gating in a Mamba-3 Block under Matched-Noise Cross-Condition Drift

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Abstract

The robustness of Mamba-style selective state-space models (SSMs) is often attributed to input-dependent selectivity. We test this attribution with a three-step ablation chain on one controlled testbed: a Mamba-3 block, bearing vibration signals (XJTU-SY, one speed-and-load transition run in both directions), and noise-aware training at matched signal-to-noise ratio. Decision thresholds for each step were written down before the corresponding grid was run; for the last grids they were first committed to a public repository. Values are macro-F1 at the final epoch of a fixed 50-epoch budget, with best-epoch values as a sensitivity analysis. First, replacing the input-dependent parts of Δ , A , B and C by learned constants costs at most about 4 pp under deep noise (paired intervals include zero). Freezing the remaining input-dependent terms of the Mamba-3 scan as well, so that no tensor entering the scan depends on the input, leaves an advantage over a parameter-matched time-invariant S4D of +21.9, +24.0 and +21.2 pp at 0, -2 and -6 dB (5/5 seeds, intervals excluding zero), with fewer parameters. Second, removing the multiplicative SiLU gate from the frozen block costs 9–23 pp and returns it to roughly the level of S4D. Third, replacing the gate in place by an additive branch with identical parameters and initial weights rescues none of that loss: the additive and gateless arms coincide at every level in both directions (differences within ± 0.6 pp, all intervals including zero; rescue ratios 0.02 and 0.01, pre-specified verdict: not rescued). Adding a gate to S4D, by contrast, recovers no more than an additive branch of

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the same size on the original transition (30% versus 41% of the gap) and only a small, direction-dependent increment on the reverse transition. The input-independent scan’s advantage and the cost of gate removal replicate on the reverse transition (+22 to +26 pp and 18–24 pp). Within this block the gate is necessary as a multiplicative operator, and what a time-invariant host gains from an added gate is largely the added structure. A separate per-class grid shows the gate-related differences concentrating on the minority inner-race class. The evidence is conditional on one block, one pair of operating conditions run in both directions with the same eight bearings, five seeds and matched-noise training.

Keywords: state-space models, Mamba, input selectivity, gating, robustness, ablation, distribution shift, matched-noise training

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet recent evidence questions it: a systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds the *simpler, time-invariant* S4D consistently ahead of Mamba-family variants in accuracy and efficiency [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (up to 18 pp), the evaluation labels are identical across all 865 cells (hash-verified), and the block can be dissected.

Our approach is deliberately *attributional* rather than *methodological*: we propose no new architecture and claim no state-of-the-art. Instead we take a regime in which a Mamba-style block exhibits a large, stable robustness advantage over a time-invariant SSM, and ask which ingredient earns it.

24 The chain has three steps, each with adjudication thresholds fixed before the
 25 corresponding grid was run:

- 26 1. **Test the classical selectivity terms (necessity test).** We con-
 27 struct a *frozen* arm: the input-dependent parts of Δ , A , B and C
 28 are replaced by learned constants. The Mamba-3 scan also takes an
 29 input-dependent trapezoidal weight and input-dependent rotation an-
 30 gles; these, the convolutional stem, the multiplicative SiLU gate, the s-
 31 tate parameterisation, normalisation and projections are retained (Sec-
 32 tion 3.2). If these four terms drive robustness, the frozen arm should
 33 collapse toward the time-invariant baseline.
- 34 2. **Localise the necessary component.** On the frozen base, we remove
 35 one component at a time (convolutional stem; SiLU gate; heavy-tailed
 36 state parameterisation) under a single-variable discipline checked key-
 37 by-key on the state dictionary (Section 3.2), and adjudicate each arm
 38 against pre-specified Δ -thresholds.
- 39 3. **Test sufficiency by constructive grafting.** The component iden-
 40 tified as necessary is added to plain S4D as a gate branch, and a pre-
 41 specified recovery ratio decides whether “removal collapses” is matched
 42 by “addition restores”.

43 The outcome is a conditional pattern (final-epoch reporting; best-epoch
 44 values in Section 4.9). Freezing the input-dependent Δ , A , B , C projections
 45 costs at most about 4 pp in deep noise (paired intervals include zero) while the
 46 block retains a 15–20 pp advantage over S4D: under the pre-specified criterion
 47 these projections are not the primary contributor. Freezing the trapezoidal
 48 weight and the rotation angles as well—so that nothing entering the scan
 49 depends on the input—does not shrink the advantage either: it measures
 50 +21 to +24 pp over S4D with fewer parameters (Section 4.2). Removing the
 51 SiLU gate costs 9–23 pp and takes the block to roughly the level of plain
 52 S4D. Adding a gate branch to S4D recovers only $R = 0.20$ of the gap (0.17
 53 under best-epoch reporting). Rebuilt in the donor’s width and projection
 54 structure, the graft recovers $R = 0.30$ (interval 0.16–0.52), but an additive
 55 branch with identical parameters recovers at least as much (0.41, Section 4.5).
 56 The converse test closes the argument: replacing the gate inside the Mamba-
 57 3 block by an additive branch with identical parameters and initial weights
 58 rescues none of the loss on either transition (rescue ratios 0.02 and 0.01,
 59 Section 4.6). The gate is necessary as a multiplicative operator in its native

block; in the S4D host it adds nothing beyond an additive branch of the same size on the original transition and a small increment on the reverse one. The first two steps replicate on the reverse transition (Section 4.7). A per-class analysis on a separate, shorter grid indicates that the gate-related differences concentrate on the minority class.

Contributions. (i) A layered freezing experiment on a Mamba-3 block under distribution drift and matched noise, from the classical selectivity terms (Δ , A , B , C) through to a scan in which no tensor depends on the input, in which the block’s deep-noise advantage over S4D survives every step; each arm is verified input-independent by capturing the tensors the scan receives for two different inputs. (ii) Component-level evidence that the output gate is the one ablated component whose removal causes a large, one-signed loss, and that adding it to a time-invariant host, even in the donor’s width and with an additive control of identical size, recovers no more than the added structure does, while replacing it in its own block by an additive branch of identical size recovers nothing on either transition; the first two findings also replicate on the reverse transition. (iii) Final-epoch reporting under a fixed budget for every arm of the chain, with best-epoch selection on the evaluation condition as a sensitivity analysis: absolute levels shift by 0.6–10.6 pp while no contrast changes sign. (iv) A documented pre-specification and review trail—evaluation-label hashes, thresholds written before each grid, and agent-assisted review—including the incidents in which automated adjudication departed from the pre-specified criteria; we release the trail.

2. Background and Related Work

2.1. Time-invariant and selective SSMs

Diagonal structured SSMs (S4D) [6] evolve a linear state with input-independent parameters (A , B , C , Δ), acting as a bank of learned linear filters. Mamba-style blocks [1] make Δ , B , C functions of the input token (selectivity), wrap the scan in a block containing a short causal convolutional stem, a multiplicative SiLU gate on a parallel branch, and residual/normalisation structure; the concrete block used throughout this paper follows the Mamba-3 formulation [7]. The narrative crediting selectivity for robustness draws on theory showing input-dependent dynamics can represent discontinuity-sensitive projections and counteract memory decay [8], and on

[9]. We treat these as *hypotheses to be tested*, not established attribution: the theoretical results concern representability, which does not by itself determine which component earns empirical robustness in a trained block.

2.2. The unresolved comparison

Empirically the picture is contested. A systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds S4D-based models superior to Mamba variants [4]; that study does not include additive noise or a shift of operating condition between training and evaluation, which is the regime examined here; theoretical work identifies long-range regimes where Mamba is weaker than S4D [5]. Meanwhile, application literatures—including machine-condition monitoring—contain many Mamba-hybrid methods reporting noise robustness [10, 11, 12, 13, 14], but none, to our knowledge, isolates *which component* of the block is responsible under controlled conditions. Our clean-data results agree with the S4D-superiority finding (S4D matches or slightly exceeds the Mamba-style block on clean data). Whether the disagreement in the literature reflects a regime difference is not settled by a single testbed; what we can report is that, in the drift $\times$ noise regime studied here, the Mamba-style block’s advantage over S4D is not removed by freezing the classical selectivity terms.

2.3. Testbed rationale

We use bearing vibration diagnosis under cross-condition drift (speed and load both change) as the testbed, for three properties that matter to the attribution question rather than to the application: (a) the distribution shift is physically generated (a change of rotation speed and radial load between recorded conditions), not simulated; (b) fault signatures are sparse transients—precisely the discontinuous structures selectivity theory predicts selective SSMs should excel at [8], which makes it a plausible arena for the selectivity hypothesis; and (c) the robustness gap between the Mamba-style block and S4D is large (up to 20 pp), giving the attribution chain high signal. The testbed is two-class (details in Section 3.1); the scope of all claims is bounded accordingly (Section 6).

3. Experimental Protocol

3.1. Testbed, arms, and budget

Vibration windows from the XJTU-SY run-to-failure dataset [15] are split by operating condition: models train on one operating condition (37.5 Hz,

11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change (cross-condition), with additive white Gaussian noise applied at matched SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The task is binary fault classification (inner-race vs. outer-race dominant classes; class imbalance makes IR the hard class, Section 4.10). All arms share an identical training budget (50 epochs, 5 seeds; the per-class grid of Section 4.10 used 20 epochs), identical splits, and identical evaluation labels, verified by a hash of the label vector: every reported cell carries `eval_sha = 6c20b367...` (Appendix A), recomputed independently during adjudication. The hash certifies the label vector, not the input windows. In total the study comprises 865 cells across five pre-specified grids (baseline grid 475; frozen-selectivity 105; component ablation 175; graft 35; per-class 75), and four follow-up grids add 255 cells (Sections 4.2, 4.9, 4.5, 4.6 and 4.7).

Composition of the data. Training uses the bearings of the 37.5 Hz/11 kN condition whose failure mode is inner race or outer race (Bearing2_1 is labelled IR; Bearing2_2, 2.4 and 2.5 are labelled OR); evaluation uses those of the 40 Hz/10 kN condition (Bearing3_3 and 3.4 are IR; Bearing3_1 and 3.5 are OR). Bearings with cage or mixed failure modes are excluded. Windows are non-overlapping segments of 2048 samples of the horizontal channel taken from the fault phase, which is delimited by a kurtosis/RMS onset criterion with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR windows, the training set 4,624 OR and 624 IR windows. Two consequences matter for interpretation. The IR class is learned from a single training bearing, so class-level results may reflect bearing-specific generalisation. And the five seeds resample initialisation, batch order and noise but not bearings: there are four independent bearings on each side, and none of the intervals reported here captures bearing-level uncertainty. One follow-up grid (Section 4.7) reverses the transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). It uses the same eight bearings with their roles swapped, so it tests the other direction rather than independent bearings; its evaluation set holds 4,624 OR and 624 IR windows, all IR windows coming from Bearing2_1, and its label hash is 93424834.... The 35 Hz/12 kN condition has no inner-race bearing and cannot form a binary pair.

The principal arms are summarised in Table 1.

3.2. What the frozen arm freezes, and what the checks verify

In the Mamba-3 layer used here, one input projection emits the gate and value streams, the scan inputs B and C , the step-size and decay terms (Δ

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Section 4.3)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Section 4.3)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Section 4.3)	112,410
frozen-trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen-rot	Frozen base with the rotation angles also made learned constants	104,346
frozen-all	Frozen base with both also constant: no tensor entering the scan depends on the input	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Section 4.4)	116,802
S4D+gate (wm)	Graft rebuilt in the donor’s shape: per layer x, z projected to width 128, S4D at width 128 (state 64, same SSM budget as the host), $W_{\text{out}}(y \odot \text{SiLU}(z))$ (Section 4.5)	198,978
S4D+branch	Same modules, parameters and initial weights, combined additively: $W_{\text{out}}(y + \text{SiLU}(z))$ (control)	198,978
frozen+add	BM3-frozen with the gate replaced in place: $W_{\text{out}}(y + \text{SiLU}(z))$ instead of $W_{\text{out}}(y \odot \text{SiLU}(z))$; same modules ⁷ parameters and initial weights (Section 4.6)	112,410
CNN/TCN baselines	Conventional baselines for context	—

and A , through input-dependent additive terms), a trapezoidal weight and rotation angles. The frozen arm removes the input-dependent parts of Δ , A , B and C and replaces them with learned constants broadcast over time (for Δ the static bias of the original layer is kept; for A a new constant parameter plays the role of a static log-decay, passed through the layer’s own heavy-tailed clamp). The removed projections are deleted, so no parameter or gradient can leak. **The trapezoidal weight and the rotation angles remain functions of the current input**, as do the stem and the output gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3 formulation [7] the input-dependent rotation and trapezoidal weighting also enter the state update. Section 4.1 is accordingly a statement about the classical selectivity terms Δ, A, B, C only. Three further arms close that gap by the same device: *frozen-trap* replaces the trapezoidal weight by a learned per-head constant, *frozen-rot* replaces the rotation angles by learned constants (so the rotation becomes a learned position-only phase rather than a function of the current input), and *frozen-all* does both, leaving **no tensor entering the scan dependent on the input**. In each case the corresponding `in_proj` output slice is deleted. Section 4.2 reports them. For these arms the construction check adds the decisive probe: the tensors handed to the scan are captured for two different inputs through the same module, and a tensor the arm claims to have frozen is required to be bitwise identical across them while a tensor left selective is required to differ (all three arms pass). No kernel modification is needed, because the scan interface accepts these tensors regardless of their origin.

Single-variable discipline was checked by key-level state-dictionary diffs between two models built under the same seed. For the frozen arm the diff verifies that every non-target tensor has the same name and shape as in BM3; for the component-ablation and graft arms it additionally compares the values of non-target tensors at initialisation. Every differing key is mapped to an entry in an architecture map. We do not claim byte-level identity of non-target tensors in the frozen arm, and trained weights naturally differ between arms; the checks concern how the arms were constructed.

3.3. Reporting rule

Every cell is trained for a fixed budget of 50 epochs with a cosine learning-rate schedule that anneals to zero, and the macro-F1 reported in the main text is that of the *final epoch*, a rule that uses no evaluation data to choose a model. The original grids recorded instead the maximum over epochs of

macro-F1 on the evaluation split (the harness defines no separate validation split), and the thresholds pre-specified for those grids were written against that rule. Every arm of the chain was therefore re-run on the adjudication band (0, -2, -6 dB) with the original training code, the same evaluation fingerprint and per-epoch evaluation (Section 4.9). Adjudications are reported under the rule they were specified for and under the final-epoch rule, and best-epoch values are given as a sensitivity analysis. Levels outside the band (clean, +10, +6, -10 dB) and the per-class grid were not re-run; values there are best-epoch and labelled auxiliary. Because the final epoch of the same evaluation condition has been inspected repeatedly during this study, final-epoch values are not a fresh confirmatory test either.

3.4. Confound elimination

Before component attribution, three candidate confounds were examined on the same grid. **Capacity:** S4D-wide ($1.75\times$ parameters) tracks S4D (S4D-wide - S4D: -0.4, -1.0, +0.4 pp at 0, -2, -6 dB under final-epoch reporting, all intervals including zero), while BM3-frozen exceeds S4D-wide by +15.9, +20.8, +16.0 pp with intervals excluding zero at all three levels, so this particular increase in S4D’s parameter count does not explain the gap; the control does not address the change in channel mixing that gate removal also entails. **Normalisation:** both SSM arms use LayerNorm and contain no BatchNorm; the arm that degrades under noise is the one *without* any batch statistics, which argues against BatchNorm statistics mismatch as the explanation. Other normalisation inside the Mamba-3 layer was not varied. **Auxiliary physics loss:** an exploratory kinematics-alignment regulariser carried by the original model contributes -0.1 pp pointwise (curves coincide); all attribution arms are therefore built on the pure cross-entropy variant.

3.5. Pre-specification and adjudication

Every adjudicated question in this study had numeric decision thresholds written to a task card and to disk before the corresponding grid was launched. This is internal pre-specification, not external registration: the supporting evidence is local file modification times and run-directory names (released, Appendix A), which are not tamper-evident and are not preserved by a repository checkout, so the trail lets readers inspect the order of events but cannot prove it. The exception is the last three follow-up grids (Sections 4.5–4.9): their thresholds, and for the last two grids the title decision that depends on their outcome, were committed to the public

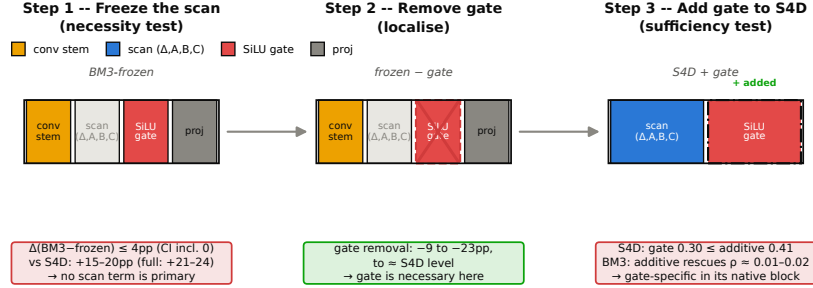
repository before the first cell was run, which gives those grids an external
 time-stamp. Later questions (component ablation, graft) were specified after
 earlier grids had been seen, as is natural in a staged design. Adjudication
 was mechanical: an analysis pass applies the thresholds and emits a decision
 artifact; an independent reviewer pass re-computes aggregates, verifies finger-
 prints row-by-row, and must cite file paths and modification times for every
 check. The pre-specified thresholds were binding in practice: two attempted
 post-hoc criterion substitutions and one stale-artifact false verification were
 caught and re-adjudicated under the original rules — this is the one place
 where pre-specification actually bit. [Appendix C](#) details the incidents, the
 automated executor/reviewer implementation, and the human adjudication
 gates; all scientific decisions — thresholds, rulings, and interpretations —
 were made by the authors and are timestamped. Statistical tests (paired,
 per-condition) are reported throughout but were never adjudication criteria.
 A single repeat of one 2-epoch configuration in a separate process differed
 by ~ 0.7 pp macro-F1 (non-deterministic kernel reductions; [Appendix B](#));
 this is an observed repeat difference, not an estimated variability bound, and
 arm-level contrasts are judged against seed-level variability (2–9 pp).

4. Results

Throughout, “deep noise” denotes the pre-specified adjudication band
 $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported but excluded from adjudication (the time-
 invariant baseline is near chance there: a constant majority-class predictor
 scores 40.0% and a uniformly random predictor 48.6% macro-F1 on this
 evaluation set, against 52.3% for S4D at the best epoch). All values are
 macro-F1 (%), mean over 5 seeds, at the final epoch of the 50-epoch budget
 (Section 3.3); best-epoch values are given alongside or in Section 4.9, and
 values outside the adjudication band are best-epoch and auxiliary. Figure 1
 summarises the three-step ablation chain that structures this section.

4.1. Freezing the input-dependent Δ , A , B , C projections

Freezing the input-dependent Δ , A , B , C projections (Section 3.2) costs
 at most 3.7 pp anywhere in the adjudication band on the point estimates
 under final-epoch reporting (+3.7, +0.9, +0.2 pp at 0, -2 , -6 dB), while
 the frozen block retains an advantage of +15.5, +19.8, +16.4 pp over S4D
 (Table 2, Figure 2). Five seeds bound these quantities loosely: paired 95% in-
 tervals for $\Delta(\text{BM3-frozen})$ are $[-3.8, 11.3]$, $[-4.5, 6.3]$, $[-12.1, 12.6]$ pp (BM3



Conditional on this block, testbed and budget: the gate is necessary as a multiplicative operator in its native block.

Figure 1: The three-step ablation chain with pre-specified thresholds: freeze the input-dependent Δ, A, B, C projections (Section 4.1), remove one component at a time on the frozen base to localise the necessary ingredient (Section 4.3), and graft that ingredient onto S4D to test sufficiency (Section 4.4).

Table 2: BM3, BM3-frozen and S4D, macro-F1 (%), mean over 5 seeds; Δ columns are seed-paired mean differences in percentage points. Upper block: final epoch of the 50-epoch budget on the adjudication band (primary; re-runs of Sections 4.2 and 4.9). Middle block: best epoch on the evaluation condition in the same re-runs (sensitivity). Lower block: best epoch in the original frozen-selectivity grid for levels outside the band (auxiliary; not re-run).

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3-frozen})$	$\Delta(\text{frozen-S4D})$
<i>Final epoch (primary)</i>					
0 dB	86.9	83.1	67.6	+3.7	+ 15.5
-2 dB	88.3	87.3	67.5	+0.9	+ 19.8
-6 dB	83.3	83.1	66.7	+0.2	+ 16.4
<i>Best epoch (sensitivity, same re-runs)</i>					
0 dB	95.6	90.6	75.9	+5.1	+14.6
-2 dB	95.8	92.3	75.1	+3.5	+17.2
-6 dB	88.6	85.7	68.6	+3.0	+17.1
<i>Auxiliary (best epoch, original grid)</i>					
clean	78.9	78.9	81.0	-0.0	-2.1
-10 dB*	68.1	64.3	52.3	+3.8	+12.0

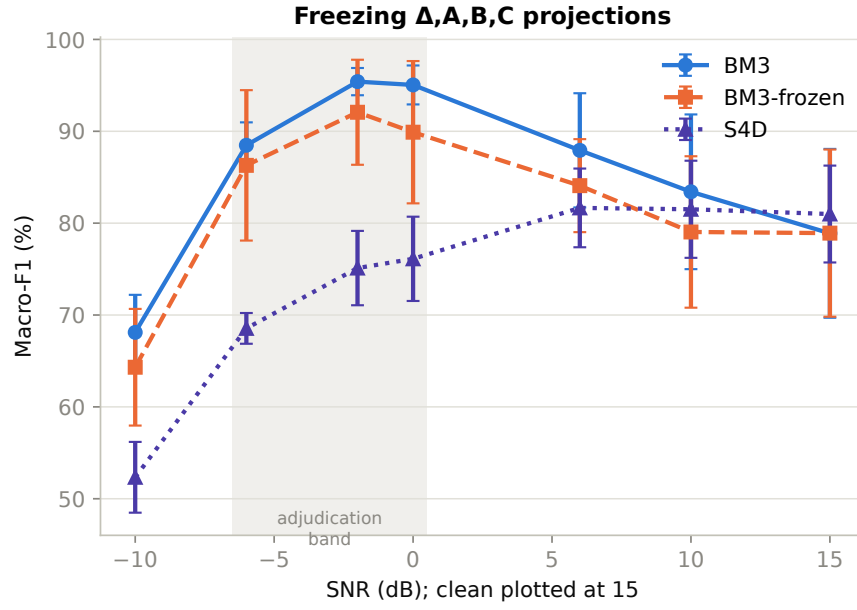


Figure 2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid (best epoch, original frozen-selectivity grid; final-epoch values on the adjudication band are in Table 2, and levels outside the band are auxiliary). The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band.

275 ahead in 4/5, 3/5, 2/5 seeds), and for $\Delta(\text{frozen-S4D})$ [4.3, 26.8], [12.0, 27.7],
 276 [4.1, 28.7] pp (5/5, 5/5, 4/5 seeds). The pre-specified rule attributed robust-
 277 ness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in deep noise with the
 278 frozen arm collapsing toward S4D. It was written for best-epoch reporting,
 279 under which the original grid gives $\Delta(\text{BM3-frozen}) = +5.1, +3.3, +2.2$ pp
 280 (re-run: +5.1, +3.5, +3.0) and $\Delta(\text{frozen-S4D}) = +13.8, +17.0, +17.7$ pp.
 281 Neither condition holds under either rule, so the pre-specified verdict is that
 282 these projections are *not the primary mechanism*. This is a failure to meet
 283 a confirmation criterion, not a demonstration that freezing is cheap: the
 284 intervals above are compatible with differences larger than 10 pp, and no e-
 285 quivalence margin was pre-specified or tested. One limit applies: the frozen
 286 arm still contains input-dependent trapezoidal weights and rotation angles, so
 287 this result concerns Δ, A, B, C only; Section 4.2 removes that limit. On clean
 288 data (auxiliary, best epoch) S4D matches or slightly exceeds both Mamba-
 289 style arms—in line with the clean-benchmark comparison of [4], which, like
 290 this auxiliary row, involves no added noise. In the noisy regime, the block’s
 291 advantage over S4D is not removed by freezing these four terms, so it must
 292 be carried by other parts of the block or by the remaining input-dependent
 293 scan terms.

294 4.2. Freezing the remaining input-dependent scan terms

295 Table 3 reports the three layered arms of Section 3.2 on the adjudication
 296 band (60 cells: 4 arms $\times$ 3 levels $\times$ 5 seeds $\times$ 50 epochs, the original training
 297 code, the same evaluation fingerprint, with the original BM3 arm re-run
 298 in the same grid). The question is whether the advantage that survived
 299 freezing Δ, A, B, C is carried by the two input-dependent terms the frozen
 300 arm retained.

301 It is not. With every tensor entering the scan input-independent, *frozen-all*
 302 is +21.9, +24.0 and +21.2 pp above S4D at 0, -2 and -6 dB under final-
 303 epoch reporting, positive in 5/5 seeds at every level and with intervals ex-
 304 cluding zero—if anything a larger margin than the partially frozen arm’s
 305 +15.5, +19.8, +16.4 pp (Table 11). None of the three freezing steps costs
 306 accuracy: the paired differences BM3-frozen $-$ frozen-trap, $-$ frozen-rot
 307 and $-$ frozen-all are negative at all three levels (between -1.8 and -6.3 pp
 308 under final-epoch reporting), i.e. the more heavily frozen arm is numerically
 309 ahead, with every interval including zero. The direction is not an artefact of
 310 capacity: *frozen-all* has 103,842 parameters against 112,410 for BM3-frozen
 311 and 177,938 for BM3. At the deepest level *frozen-all* also exceeds the full

Table 3: Layered freezing of the scan’s input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t -intervals over seeds.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen-trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen-rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen-all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
<i>Paired differences, final-epoch reporting (pp)</i>						
frozen-all − S4D	+21.9 [16.7, 27.1]		+24.0 [21.4, 26.5]		+21.2 [17.5, 24.9]	
BM3-frozen − frozen-all	−6.3 [−17.6, 4.9]		−4.1 [−12.0, 3.7]		−4.8 [−17.8, 8.2]	
BM3 − frozen-all	−2.6 [−8.2, 2.9]		−3.2 [−7.0, 0.6]		−4.5 [−7.2, −1.9]	

BM3 block by 4.5 pp ($[-7.2, -1.9]$ for BM3 − frozen-all, 0/5 seeds in BM3’s favour); we report this but do not build on it, since five seeds and one speed pair cannot separate a genuine advantage of a time-invariant scan from a variance effect.

Two readings of the whole step are therefore available, and the data do not separate them: input-dependence of the scan contributes nothing measurable here, or it contributes something that the learned constants recover under this training budget. Either way the conclusion for the attribution chain is the same and now rests on a complete intervention rather than a partial one: **the Mamba-style block’s deep-noise advantage over a time-invariant S4D does not require any input-dependent term in its scan**. What remains to carry it is the rest of the block—the convolutional stem, the heavy-tailed A parameterisation, the normalisations and the output gate—which is what Sections 4.3–4.4 examine. The re-run also reproduces the original grids: the BM3 arm’s best-epoch means (95.6, 95.8, 88.6) exceed the original frozen-selectivity grid’s 95.0, 95.4, 88.5 by 0.57, 0.43 and 0.16 pp (a different PyTorch/Triton build, Section 4.9).

4.3. Removing the gate is the only ablation with a large, one-signed effect

“Necessary” is used here in the operational sense of this ablation: removing the gate from the frozen block, with the other components and the budget unchanged, removes most of the block’s advantage. The gate result is large,

Table 4: Single-component removal on the frozen base, $\Delta(\text{frozen} - \text{arm})$ in percentage points: final-epoch mean, 95% t -interval over seeds and number of seeds (of 5) with a positive difference (re-runs of Section 4.9); “best” is the best-epoch value of the original component-ablation grid, for which the thresholds were pre-specified.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
SiLU gate	+9.1 [2.9, 15.3] (5/5); best +13.1	+15.6 [−0.8, 32.0] (4/5); best +21.1	+22.6 [5.8, 39.4] (5/5); best +23.5	PRIMARY (all $\geq$ 8 pp, both rules)
conv stem ($k=7$)	−13.2 [−24.1, −2.2] (0/5); best −6.6	−7.5 [−14.6, −0.3] (0/5); best −2.7	+5.9 [−4.0, 15.8] (3/5); best +7.9	outside bins; regime trade-off
heavy-tailed A $\rightarrow$ S4D-style A	−1.9 [−9.2, 5.3] (2/5); best −2.4	−2.6 [−8.6, 3.4] (2/5); best −2.2	−4.8 [−16.4, 6.9] (2/5); best −4.2	outside bins; inter- vals include 0

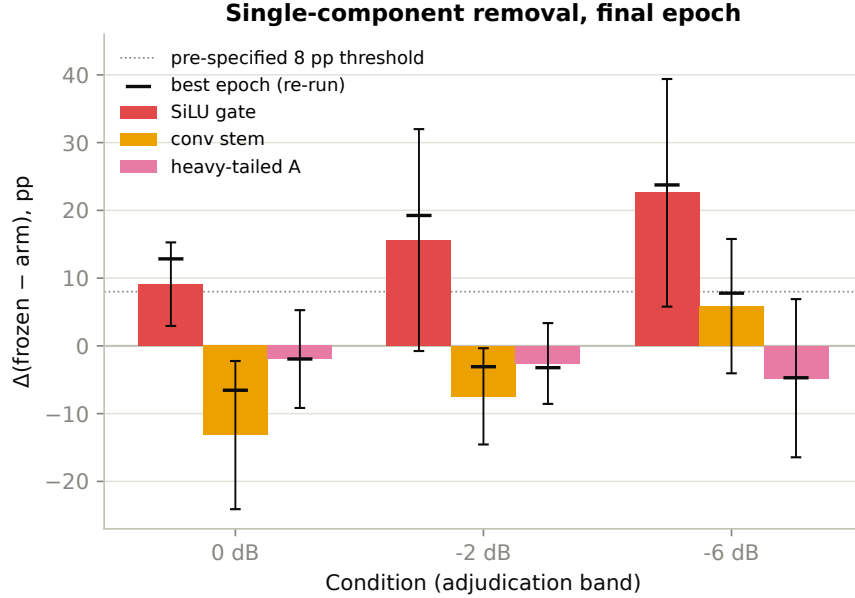


Figure 3: Component-removal Δ (percentage points, frozen $-$ arm) on the adjudication band for the three single-component ablations on the frozen base: final-epoch means with 95% t -intervals over seeds (bars) and best-epoch means of the same re-runs (black ticks). Only gate removal is large and one-signed at every level and clears the pre-specified 8 pp threshold.

one-signed and grows with noise depth: under final-epoch reporting removal
 costs 9.1, 15.6, 22.6 pp at 0, -2 and -6 dB (paired 95% intervals [2.9, 15.3],
 $[-0.8, 32.0]$, [5.8, 39.4]; 5/5, 4/5, 5/5 seeds), above the pre-specified 8 pp
 threshold at every level; the interval at -2 dB includes zero because of the
 large seed variance of the gateless arm. Under the best-epoch rule for which
 the threshold was written, the component-ablation grid gives 13.1, 21.1 and
 23.5 pp ([6.5, 19.6], [3.8, 38.5], [8.5, 38.6]; 5/5 seeds at every level), so the ver-
 dict is *primary* under both rules. The gateless arm gives up most of the
 Mamba advantage but is not resolved against plain S4D: frozen-gate - S4D
 is +6.4, +4.2, -6.2 pp, with every interval including zero. We therefore de-
 scribe it as roughly at the level of S4D, below it only on the point estimate at
 -6 dB; this suggests, but does not establish, that components interact rather
 than add. The other two components tell regime stories rather than robust-
 ness stories. Removing the convolutional stem is beneficial at 0 and -2 dB
 (frozen - frozen-conv = -13.2 [$-24.1, -2.2$] and -7.5 [$-14.6, -0.3$], with
 the stemless arm ahead in every seed) and costly at -6 dB ($+5.9$, [$-4.0, 15.8$],
 3/5 seeds); under best-epoch reporting the pattern was the same (-6.6 , -2.7 ,
 $+7.9$ pp). On clean cross-speed data (auxiliary, best epoch, original grid) the
 stem costs 21.0 pp (removal yields 98.8 vs. 77.8, with seed-std collapsing
 from 8.1 to 0.6 pp)—consistent either with speed-specific local temporal pat-
 terns mistransferring across conditions, or with optimisation friction under
 the matched budget; we report both readings. Replacing the heavy-tailed
 state parameterisation by an S4D-style one is numerically beneficial at all
 three levels (by 1.9, 2.6, 4.8 pp under final-epoch and 2.4, 2.2, 4.2 pp under
 best-epoch reporting), but every paired interval includes zero, so we draw no
 directional conclusion from it.

4.4. Adding a narrow gate branch to S4D

The pre-specified recovery ratio is $R = \text{mean over the band of (S4D+gate-
 S4D)/(frozen - S4D)}$. Under final-epoch reporting $R = \mathbf{0.196}$ (per level
 0.249, 0.341, -0.003 ; seed-resampled 95% interval [0.09, 0.31]); under the
 best-epoch rule for which it was specified, $R = 0.174$ ([0.04, 0.29]). The
 pre-specified bins were $R \geq 0.7$ (confirmed), $0.3 \leq R < 0.7$ (partial recov-
 ery) and $R < 0.3$ (insufficient); both values fall in the last, although the
 final-epoch interval reaches the 0.3 boundary. The per-level paired gains of
 S4D+gate over S4D under final-epoch reporting are +3.9 ($[-0.8, 8.6]$), +6.8
 ([2.4, 11.1]), -0.1 ($[-4.0, 3.9]$) pp. At the deepest adjudicated level the graft
 recovers nothing. Removal and addition are not mirror-image interventions.

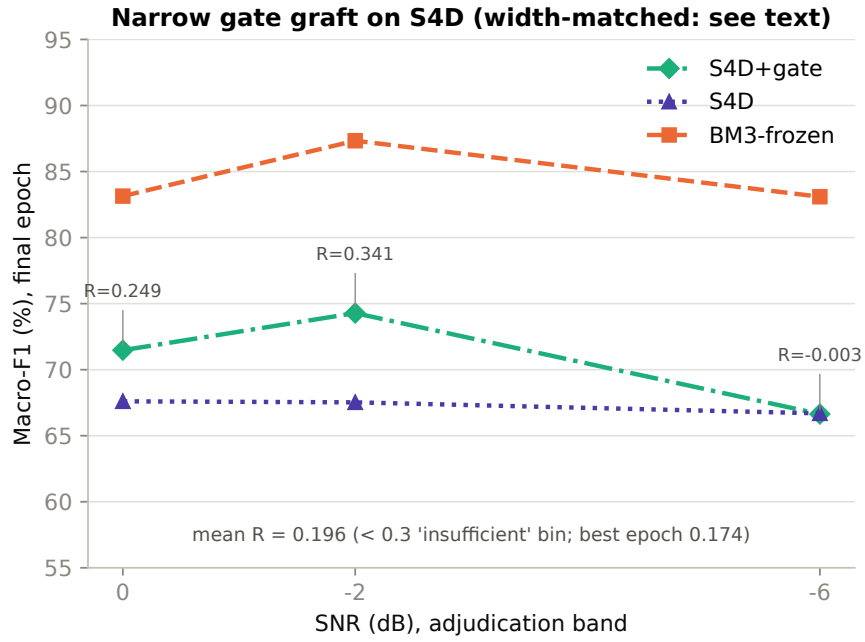


Figure 4: Graft outcome under final-epoch reporting: S4D+gate, plain S4D, and BM3-frozen on the adjudication band (one re-run grid), with the per-level recovery ratio $R = (S4D+gate - S4D) / (frozen - S4D)$ annotated. Recovery is near zero at the deepest adjudicated noise level. This is the narrow graft; the width-matched graft and its additive control are in Table 6.

Table 5: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added). Upper block: final epoch, all three arms from the same re-run (Section 4.9). Lower block: best epoch as pre-specified, with S4D+gate from the graft grid and S4D and BM3-frozen, and hence R , from the component-ablation grid of Section 4.3 (independent runs of the same arms differ by 0.1–0.7 pp).

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
<i>Final epoch (primary)</i>				
0 dB	71.5	67.6	83.1	0.249
−2 dB	74.3	67.5	87.3	0.341
−6 dB	66.6	66.7	83.1	−0.003
<i>Best epoch (as pre-specified)</i>				
0 dB	79.8	76.2	90.6	0.249
−2 dB	79.4	75.2	92.8	0.239
−6 dB	69.2	68.6	86.0	0.035

370 In BM3 the gate has its own $64 \rightarrow 128$ projection and multiplies the scan
371 output at the inner width of 128 before the output projection; removing it
372 deletes 32,768 parameters. In the S4D host the grafted gate is a $64 \rightarrow 64$
373 linear branch (with bias) computed from the layer input and multiplying the
374 SSM output before the residual addition; it adds 16,640 parameters over four
375 layers. The two share the multiplicative SiLU form but not width, projec-
376 tion structure or parameter count. Removal costs 9–23 pp in the frozen block,
377 whereas this graft changes S4D by between −0.1 and +6.8 pp (final epoch).
378 What this shows is that *this gate branch, in this host and under this budget,*
379 *recovers little*. Three readings fit the data and cannot be separated here: (i)
380 host dependence, i.e. the gate needs the stem or scan context it co-trained
381 with; (ii) implementation differences in width, projection and placement; (iii)
382 transplant friction, i.e. the grafted branch is harder to optimise in a foreign
383 host under the matched budget (training converges normally, but convergence
384 is not robustness). Section 4.5 tests reading (ii) with a width-matched graft
385 and an equal-capacity non-multiplicative control; a tuning or longer-budget
386 comparison selected on a source-domain validation set was not run.

387 The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) -$
388 $F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under
389 last-epoch reporting $I = 5.2$ [2.3, 8.1], 8.9 [−6.2, 23.9] and 22.6 [2.9, 42.4] pp
390 at 0, −2 and −6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and

5/5 seeds); under best-epoch reporting in the re-run it is 7.8 [1.4, 14.2], 14.4 [−1.4, 30.3] and 23.1 [8.7, 37.4] pp. A positive I means the gate adds more on the frozen host than on S4D. Because the host differs in several respects at once and the two gates are not aligned, I cannot be assigned to a specific component or interaction term, and the seed pairing across arms is nominal (arms share seed indices, not initialisations).

4.5. A width-matched graft and an additive control

The graft of Section 4.4 is narrower and smaller than the donor gate. Two further arms rebuild it in the donor’s shape (Table 1). In each layer the layer input is projected to x and z of width 128 (the frozen block’s inner width), the S4D layer runs on x at width 128 with state size 64—so that its own parameter count and total state size equal the host’s 64-wide, state-128 layer—and an output projection maps back to width 64. *S4D+gate (wm)* combines $y \odot \text{SiLU}(z)$, as the donor does; *S4D+branch* uses the same modules, parameters and initial weights but combines $y + \text{SiLU}(z)$. Both have 198,978 parameters, more than BM3-frozen (112,410) and BM3 (177,938), which biases the test towards recovery. A construction check with forward hooks confirms that the tensor entering the output projection equals the intended formula for each arm and not the other, and that the two arms are identical at initialisation. The thresholds were committed publicly before the grid ran (Section 3.5): R as before, under final-epoch reporting; the verdict is INCONCLUSIVE whenever the seed-resampled interval crosses a bin boundary, in which case we write at the tier less favourable to the claim; and a gate-specific reading requires $\Delta R = R_{\text{gate}} - R_{\text{branch}} > 0.1$ with an interval excluding zero.

The width-matched graft recovers $R = 0.296$ of the gap (per level 0.159, 0.298, 0.431). Its interval, [0.16, 0.52], crosses the 0.3 boundary, so the pre-specified verdict is INCONCLUSIVE between “insufficient” and “partial”, and we report it at the less favourable tier: a graft built in the donor’s shape recovers part of the gap. The recovery is not gate-specific. The additive branch, with the same parameters, recovers at least as much ($R = 0.408$), $\Delta R = -0.112$ [−0.26, 0.02], and at 0 dB the gated arm is below the additive one by 3.0 pp [−4.5, −1.5], 0/5 seeds). With frozen—all in the denominator the gated arm’s ratio is 0.231 [0.13, 0.33]; under best-epoch reporting the two ratios are 0.459 and 0.483. What a graft recovers on this transition therefore follows the added structure—a wider channel axis and input/output

Table 6: Width-matched graft and additive control on the original transition. Macro-F1 (%), mean over 5 seeds, final-epoch reporting; R is the mean over the three levels of (arm – S4D)/(BM3-frozen – S4D), with a seed-resampled 95% interval. S4D, BM3-frozen and the narrow graft are the re-runs of Section 4.9.

Arm	0 dB	–2 dB	–6 dB	R	95% interval
S4D	67.6	67.5	66.7	–	–
S4D+gate (narrow)	71.5	74.3	66.6	0.196	[0.09, 0.31]
S4D+gate (wm)	70.1	73.4	73.8	0.296	[0.16, 0.52]
S4D+branch	73.1	76.2	73.8	0.408	[0.31, 0.60]
BM3-frozen	83.1	87.3	83.1	1	–
$\Delta R = \text{S4D+gate (wm)} - \text{S4D+branch}$				–0.112	[–0.26, 0.02]

projections—rather than the multiplicative interaction. Reading (ii) of Section 4.4 is partly confirmed, in that the narrow graft understated what added structure can recover; but in the S4D host the gate itself contributes nothing beyond an equal-capacity additive branch, whereas removing it from the frozen Mamba-3 block costs 9–23 pp under the same rule (Table 11).

4.6. Replacing the gate in its native host

The previous subsection changes the host and keeps the operator. A pre-specified grid (15 cells, original transition) does the converse: it keeps the host and changes the operator. In *frozen+add* the scan kernel of every BM3-frozen layer is called without gating and the layer output becomes $W_{\text{out}}(y + \text{SiLU}(z))$; modules, parameter count (112,410) and initial weights are bitwise those of BM3-frozen, and a construction check confirms that the donor operator is $y \odot \text{SiLU}(z)$ and that the new arm computes $y + \text{SiLU}(z)$. The pre-specified statistic is the native rescue ratio $\rho = (\text{add} - \text{no gate})/(\text{frozen} - \text{no gate})$, mean over levels, final epoch, with a seed-resampled interval; *not rescued* requires $\rho < 0.3$ with an upper bound below 0.5.

The additive branch rescues nothing. *frozen+add* is indistinguishable from the gateless arm at every level (+0.6, –0.2, +0.4 pp, every interval including zero), and the multiplicative arm exceeds it by 8.5, 15.8, 22.2 pp. The rescue ratio is $\rho = 0.024$ (per level 0.070, –0.015, 0.016), with a seed-resampled 95% interval of [–1.159, 0.486]; under best-epoch reporting $\rho = 0.005$. By the pre-specified rule the verdict is *not rescued*. Two features of the interval should be read with it: its upper bound lies only 0.014 below the threshold, and its lower bound is wide because the 0 dB denominator is only

Table 7: Gate replaced by an additive branch inside BM3-frozen, original and reverse transitions. Macro-F1 (%), mean over 5 seeds; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
<i>Original transition (37.5 Hz/11 kN → 40 Hz/10 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	90.6	83.1	92.3	87.3	85.7	83.1
frozen+add ($y + \text{SiLU}(z)$)	78.0	74.7	72.6	71.5	62.3	60.9
frozen−gate (no z)	77.7	74.0	73.1	71.7	61.9	60.5
BM3-frozen − frozen+add	+8.5		+15.8		+22.2	
	[1.1, 15.9] (5/5)		[−1.2, 32.9] (4/5)		[4.6, 39.9] (5/5)	
frozen+add − frozen−gate	+0.6		−0.2		+0.4	
	[−1.7, 3.0] (3/5)		[−3.4, 3.0] (3/5)		[−0.7, 1.4] (3/5)	
<i>Reverse transition (40 Hz/10 kN → 37.5 Hz/11 kN)</i>						
BM3-frozen ($y \odot \text{SiLU}(z)$)	81.1	74.7	84.0	76.7	80.8	73.9
frozen+add ($y + \text{SiLU}(z)$)	64.3	56.5	62.2	54.4	55.7	50.5
frozen−gate (no z)	63.8	56.2	62.0	54.7	57.1	50.0
BM3-frozen − frozen+add	+18.1		+22.3		+23.4	
	[1.2, 35.0] (5/5)		[3.5, 41.0] (5/5)		[6.8, 40.0] (5/5)	
frozen+add − frozen−gate	+0.3		−0.3		+0.6	
	[−0.6, 1.2] (3/5)		[−1.0, 0.4] (2/5)		[−0.2, 1.4] (5/5)	

9.1 pp (in 7.8% of resamples one level’s denominator fell below 2 pp and was excluded, as pre-specified).

A further pre-specified grid (15 cells) repeats the substitution on the reverse transition (Section 4.7), with BM3-frozen and frozen–gate taken from the reverse grid. The outcome is the same (Table 7, lower block): frozen+add coincides with the gateless arm at every level (+0.3, −0.3, +0.6 pp, every interval including zero), and the multiplicative arm exceeds it by +18.1, +22.3, +23.4 pp (intervals excluding zero, 5/5 seeds at every level). The rescue ratio is $\rho = 0.009$ (per level 0.017, −0.014, 0.023), with a seed-resampled 95% interval of $[-0.943, 0.362]$; under best-epoch reporting $\rho = -0.005$. The pre-specified verdict is again *not rescued*, here with the upper bound 0.14 below the threshold.

Together with Section 4.5 this gives the two halves of a symmetric test. In the S4D host the gate adds nothing beyond an additive branch of the same size; in the Mamba-3 host an additive branch of the same size adds nothing beyond removing the gate, on both transitions. The benefit of the gate is specific to the multiplicative operator *in its native host*, and what transfers to the time-invariant host is the added structure, not the gating.

4.7. The reverse transition

A pre-specified grid repeats the chain’s central contrasts on the reverse transition, 40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN (75 cells: frozen—all, BM3-frozen, frozen–gate, S4D and S4D+gate (wm), at three levels and five seeds; Section 3.1 for the data). This direction is harder for the time-invariant model: on its evaluation set, predicting the majority class gives a macro-F1 of 46.8% and uniform random guessing 41.5%, and S4D sits close to those levels (Table 8).

The first two steps replicate. The fully input-independent scan exceeds S4D by +26.3, +25.6 and +22.2 pp at 0, −2 and −6 dB, with intervals excluding zero and 5/5 seeds at every level, so by the pre-specified rule (at least two of three levels) the first step is replicated on the reverse transition. Removing the gate from BM3-frozen costs 18.4, 22.0 and 24.0 pp, above the 8 pp threshold at all three levels, with intervals excluding zero and 5/5 seeds, and the gateless arm is indistinguishable from S4D (intervals include zero). On the third step, which the pre-specification treats as descriptive here, the width-matched graft recovers most of the gap, $R = 0.911$ $[0.82, 1.04]$ (0.863 with frozen—all in the denominator). A further pre-specified grid (15 cells) adds the same-size additive control in this direction: it reaches 67.8, 72.1,

Table 8: Reverse transition (40 Hz/10 kN $\rightarrow$ 37.5 Hz/11 kN). Macro-F1 (%), mean over 5 seeds, best-epoch and final-epoch reporting; paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference.

Arm (best / final)	0 dB	-2 dB	-6 dB
frozen-all	86.3 / 77.5	87.8 / 77.9	81.7 / 74.0
BM3-frozen	81.1 / 74.7	84.0 / 76.7	80.8 / 73.9
S4D+gate (wm)	79.0 / 72.9	80.5 / 73.9	78.5 / 72.2
frozen-gate	63.8 / 56.2	62.0 / 54.7	57.1 / 50.0
S4D	61.2 / 51.2	61.0 / 52.4	54.6 / 51.8
	+26.3	+25.6	+22.2
frozen-all - S4D	[21.5, 31.0] (5/5)	[21.1, 30.1] (5/5)	[18.8, 25.6] (5/5)
	+18.4	+22.0	+24.0
BM3-frozen - frozen-gate	[1.2, 35.6] (5/5)	[3.4, 40.6] (5/5)	[7.6, 40.3] (5/5)
	+5.0	+2.4	-1.8
frozen-gate - S4D	[-17.6, 27.7] (2/5)	[-19.2, 23.9] (2/5)	[-18.9, 15.3] (1/5)
	+21.7	+21.5	+20.5
S4D+gate (wm) - S4D	[15.4, 28.0] (5/5)	[17.0, 26.0] (5/5)	[15.7, 25.2] (5/5)

68.4% (final epoch) and recovers $R = 0.758$ [0.66, 0.88]. The gate arm exceeds it by +5.1, +1.7, +3.8 pp (only the -6 dB interval excludes zero), and the difference in recovery is $\Delta R = 0.153$ [0.06, 0.26]; under best-epoch reporting it shrinks to 0.037 [-0.1, 0.17]. By the pre-specified descriptive labels this is *unresolved*: in this direction the gate carries a small increment over the additive branch, whereas on the original transition it carried none ($\Delta R = -0.112$). Most of the reverse-direction recovery is therefore available to an additive branch of the same size, and the remaining gate increment is direction-dependent. Both ratios rest on an S4D baseline close to the trivial floor (51–52% against 46.8% for the majority class). Both statements carry the limits of this pair: the same eight bearings with their roles swapped, and a single inner-race bearing in the evaluation set.

4.8. Replication attempt on a second dataset with disjoint bearings

A pre-specified grid (75 cells; thresholds, a ceiling-and-floor check and the writing consequences committed before the first cell) repeats the three central contrasts on the Paderborn KAt data [16], using real-damage bearings only. Models train on two outer-race (KA04, KA16) and two inner-race bearings (KI04, KI14) at 1500 rpm and are evaluated on five different bearings (KA15, KA22; KI17, KI18, KI21) at 900 rpm, with load and radial force unchanged;

507 the vibration channel is resampled to 25.6 kHz so that a window spans 80 ms
508 as in XJTU-SY, and everything else (noise, training, arms, budget) is as in
509 the main grids. The pre-specified smoke check (two arms, one extra seed)
510 passed its three criteria, the learnability criterion by 0.3 pp.

Table 9: Paderborn replication (KA04, KA16, KI04, KI14 at 1500 rpm $\rightarrow$ KA15, KA22, KI17, KI18, KI21 at 900 rpm). Upper block: macro-F1 (%), mean over 5 seeds, best / final epoch; the majority-class macro-F1 on this test set is 37.5%. Lower block: paired differences under final-epoch reporting with 95% t -intervals and the number of seeds (of 5) with a positive difference.

Arm (best / final)	0 dB	-2 dB	-6 dB
BM3-frozen	55.1 / 51.9	53.5 / 50.3	51.5 / 47.7
frozen-all	56.1 / 54.2	54.6 / 51.3	51.4 / 47.8
S4D	57.0 / 55.6	55.1 / 53.9	51.5 / 51.1
frozen-gate	57.9 / 56.9	56.5 / 54.9	52.4 / 51.1
frozen+add	58.0 / 56.9	56.5 / 54.8	52.9 / 51.0
frozen-all - S4D	-1.4 [-4.0, 1.2] (1/5)	-2.5 [-5.3, 0.2] (1/5)	-3.3 [-5.8, -0.8] (0/5)
BM3-frozen - S4D	-3.8	-3.6	-3.4
	[-11.9, 4.3] (1/5)	[-12.6, 5.4] (2/5)	[-8.9, 2.1] (1/5)
BM3-frozen - frozen-gate	-5.0	-4.6	-3.3
	[-8.8, -1.3] (0/5)	[-9.2, 0.1] (0/5)	[-7.1, 0.4] (0/5)
frozen+add - frozen-gate	0.0	0.0	-0.1
	[-0.9, 0.8] (2/5)	[-0.4, 0.3] (3/5)	[-0.4, 0.2] (1/5)

511 None of the pre-specified contrasts replicates (Table 9). The fully input-
512 independent block does not exceed S4D (frozen-all - S4D: -1.4, -2.5,
513 -3.3 pp, the interval at -6 dB excluding zero in the negative direction), nor
514 does BM3-frozen (-3.8, -3.6, -3.4 pp). Removing the gate does not cost ac-
515 curacy: BM3-frozen - frozen-gate is -5.0, -4.6, -3.3 pp, below zero in ev-
516 ery seed at every level. With no gate effect to rescue, the native-substitution
517 test is not assessable; the additive and gateless arms again coincide (0.0, 0.0,
518 -0.1 pp). The pre-specified labels are therefore *not replicated* for both freeze
519 contrasts and for gate removal, and *not assessable* for the substitution test.

520 Two observations bound how this result can be read, and neither changes
521 the labels. First, the testbed discriminates poorly: every arm ends between
522 47.7 and 56.9% macro-F1 at every level, all arms fit the training condition
523 (the gated arms reach the lowest training loss), and none transfers far above
524 the level reached after the first epoch. Second, where differences are resolved

525 they run against the Mamba-3 arms, which reach lower training loss and
526 transfer slightly worse. On this second dataset, therefore, the block’s advan-
527 tage under matched noise and drift is not observed; we do not know whether
528 this reflects a property of the PU transition (speed reduction to 900 rpm with
529 disjoint bearings), the protocol transferred unchanged from XJTU-SY, or a
530 limit of the XJTU-SY result itself, and we restrict all claims in this paper to
531 XJTU-SY.

532 4.9. Sensitivity to the reporting rule

533 The main tables use the final epoch of the 50-epoch budget (Section 3.3).
534 The original grids recorded instead the best epoch on the evaluation condi-
535 tion, and their pre-specified thresholds were written for that rule. To mea-
536 sure how much the rule shapes the conclusions, every arm of the chain was
537 re-run over the adjudication band with per-epoch evaluation, using the o-
538 riginal training code and the same evaluation fingerprint: the four arms
539 BM3-frozen, S4D, frozen–gate and S4D+gate (60 cells), the BM3 arm a-
540 longside the layered freezes (Section 4.2), and the capacity control and the
541 two other single-component removals, S4D-wide, frozen–conv and frozen–A
542 (45 cells, pre-specified with a rule for changes of sign). The re-runs used a dif-
543 ferent PyTorch/Triton build from the original grids; their best-epoch values
544 reproduce the original arm-by-condition means to within -0.6 to $+1.4$ pp
545 across the eight re-run arms (the largest deviation is in the high-variance
546 frozen–gate arm at -2 dB).

Table 10: Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting in the re-runs, for every arm of the chain on the adjudication band.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
S4D-wide	77.8	67.3	73.3	66.5	69.0	67.1
frozen–gate	77.7	74.0	73.1	71.7	61.9	60.5
frozen–conv	97.1	96.3	95.4	94.8	77.9	77.2
frozen–A	92.5	85.1	95.6	89.9	90.4	87.9
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

Best-epoch values exceed final-epoch values by 0.6–10.6 pp across the 24 arm-by-level means, so best-epoch absolute levels are optimistic. The reduction is of similar size for the arms being contrasted (e.g., BM3-frozen –7.4, –5.0, –2.6 pp and S4D –8.3, –7.6, –1.9 pp at 0, –2, –6 dB), and the contrasts that carry the argument are correspondingly stable (Table 11).

Table 11: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference.

Contrast	0 dB	–2 dB	–6 dB
	+15.5	+19.8	+16.4
BM3-frozen – S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen – frozen–gate	[2.9, 15.3] (5/5)	[–0.8, 32.0] (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	–6.2
frozen–gate – S4D	[–2.9, 15.8] (4/5)	[–14.5, 22.9] (3/5)	[–23.5, 11.2] (1/5)
	+3.9	+6.8	–0.1
S4D+gate – S4D	[–0.8, 8.6] (4/5)	[2.4, 11.1] (5/5)	[–4.0, 3.9] (2/5)

Selectivity. The frozen block’s advantage over S4D is +15.5, +19.8 and +16.4 pp (and +21.9, +24.0, +21.2 pp for the fully frozen scan, Section 4.2), versus +14.6, +17.2 and +17.1 pp under best-epoch reporting in the re-run. It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding zero at every level. *Gate.* Removing the gate costs 9.1, 15.6 and 22.6 pp (12.8 to 23.8 pp under best-epoch reporting in the re-run), above the 8 pp pre-specified threshold at every level; the interval at –2 dB includes zero ([–0.8, 32.0] pp, 4/5 seeds) because of the large seed variance of the gateless arm (standard deviation 14.6 pp at –2 dB). *Gateless versus S4D.* Under final-epoch reporting the gateless arm is not below S4D at 0 and –2 dB (point estimates +6.4 and +4.2 pp) and is below it at –6 dB (–6.2 pp); all three intervals include zero. The statement that gate removal “falls below S4D” is therefore supported only as a point-estimate observation at the deepest level, and we phrase it as “to or below” elsewhere. *Graft.* The recovery ratio is 0.249, 0.341 and –0.003 per level and $R = 0.196$ on average (best-epoch re-run: 0.220), again in the pre-specified “insufficient” bin on the point estimate. The seed-resampled 95% interval is [0.09, 0.31] and so reaches the 0.3 boundary: with five seeds the data support “far below confirmation (0.7) and probably below 0.3”, not a precise value of R . The graft gain over S4D is resolved at –2 dB only (+6.8 pp, [2.4, 11.1]) and is indistinguishable from zero at –6 dB.

572 *Remaining arms.* By the rule pre-specified for the 45-cell grid, no arm
 573 changes sign between the two rules at any level where both values exceed
 574 1 pp, so all main-table entries are reported under the final-epoch rule with
 575 unchanged wording. The capacity control holds: BM3-frozen exceeds S4D-
 576 wide by +15.9, +20.8, +16.0 pp (intervals excluding zero at all three levels),
 577 and S4D-wide – S4D is –0.4, –1.0, +0.4 pp (intervals including zero). For
 578 the stem the final-epoch rule sharpens the regime trade-off (intervals at 0
 579 and –2 dB now exclude zero) without reversing it.

580 Because the final epoch of the same evaluation condition has been in-
 581 spected repeatedly during this study, last-epoch values are also not a fresh
 582 confirmatory test. The reporting rule therefore changes absolute levels and
 583 the strength of the gateless-versus-S4D statement, but not the ordering BM3-
 584 frozen > {S4D+gate, frozen–gate, S4D}, the size of the frozen block’s ad-
 585 vantage over S4D, or the verdicts of the pre-specified gate and graft rules.

586 4.10. The interaction effect concentrates on the hard class

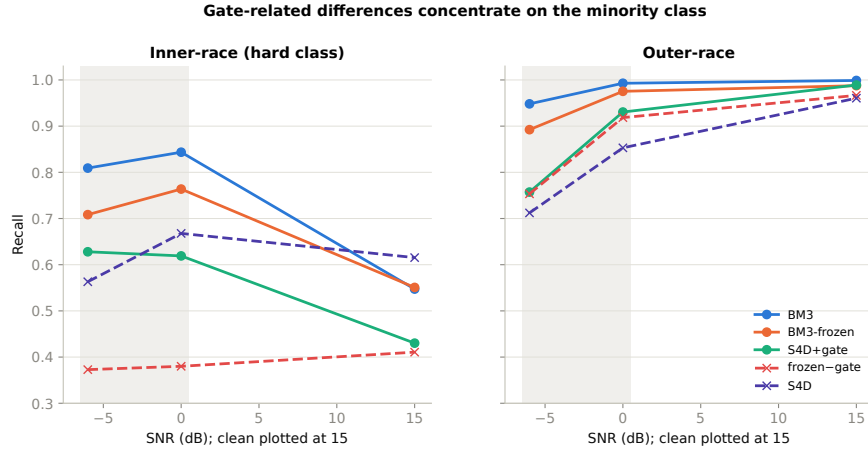


Figure 5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate.

587 Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch
 588 selection on the evaluation condition: 5 arms $\times$ {clean, 0, –6 dB} $\times$ 5 seeds,
 589 per-sample predictions dumped and re-verified against reported metrics cell-
 590 by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62
 591 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate

592 presence: in all three gated arms IR recall at -6 dB exceeds its clean val-
593 ue (e.g., $0.55 \rightarrow 0.81$ for BM3), while in both gateless arms it is below its
594 clean value (frozen-gate: $0.41 \rightarrow 0.37$; S4D: $0.62 \rightarrow 0.56$, non-monotone,
595 rising to 0.67 at 0 dB) (Figure 5). Both ablation directions hit IR hardest—
596 gate removal costs IR 2.3 – $3.8\times$ more F1 than OR at every level (F1 from
597 predictions pooled over seeds), and the graft’s noise-time gains also concen-
598 trate on IR (-6 dB: $+6.8$ pp IR vs. $+4.2$ pp OR). Because macro-F1 averages
599 the two classes, the arm-level robustness gaps of Sections 4.1–4.4 are carried
600 substantially by the hard class. The gate-related differences are therefore
601 class-concentrated rather than uniform, and fall on the class with the lowest
602 recall. We did not measure classification margins, so we do not describe IR as
603 a “low-margin” class. Because this grid is separate from the main 50-epoch
604 grid and the IR class comes from a single training bearing (Section 3.1), these
605 results are descriptive and do not decompose the main-grid effects.

606 5. Discussion

607 **What can be credited?** The ablation chain supports a narrow, con-
608 ditional statement: within this Mamba-3 block, on this testbed and with
609 noise-aware training, (i) making the scan fully input-independent—first the
610 Δ , A , B , C projections, then the trapezoidal weight and rotation angles as
611 well—did not remove the block’s deep-noise advantage over a time-invariant
612 S4D, and at the deepest level the fully frozen arm matched or exceeded the
613 original block with fewer parameters, (ii) removing the output gate removed
614 most of that advantage—(i) and (ii) holding on the reverse transition as
615 well—and (iii) adding the gate to S4D recovered part of the advantage only
616 together with added width and projections, and an additive branch of the
617 same size recovered as much, while (iv) in the Mamba-3 block an additive
618 branch of the same size recovered none of what removing the gate cost, on
619 both transitions. These observations are consistent with a gate benefit that is
620 specific to the multiplicative operator in its native host. They do not identify
621 which other component the gate depends on. Nor do they separate “input-
622 dependence contributes nothing here” from “what it contributes is recovered
623 by learned constants under this budget”; the two readings are indistinguish-
624 able in our data and lead to the same attribution. Representability results for
625 selectivity [8] are not contradicted by a single ablation on one testbed. The
626 S4D-superiority finding [4] was obtained on clean classification benchmarks,
627 without matched noise or a shift of operating condition, so it does not speak

628 to the regime studied here and we do not read it as covering it. Two points
629 of contact are nevertheless consistent with it: our clean-data parity between
630 S4D and the Mamba-style arms, and our finding that input-dependent se-
631 lectivity is not needed for the block’s advantage. What this paper adds is
632 specific to the matched-noise, cross-condition regime: there S4D trails the
633 frozen Mamba-3 block by 15–26 pp under final-epoch reporting (partial and
634 full freeze, both transitions), and the gap is attributable to the multiplicative
635 output gate in its native block rather than to selectivity. This is a statement
636 about a different regime, not a contradiction of the clean-benchmark result.

637 **Implications for backbone selection and for transplant practice.**

638 For practitioners choosing backbones for noisy, drifting signal regimes, the
639 results suggest that the block’s benefit should be judged as a composition,
640 and that a gate added to a different host should not be expected to reproduce
641 it: in our S4D host a gate built in the donor’s shape recovered no more than
642 an additive branch of the same size on the original transition (Section 4.5)
643 and only slightly more on the reverse one (Section 4.7), and neither design
644 was tuned. A hypothesis for future work follows: a recomposed block guided
645 by the component map (no conv stem, gate retained, S4D-style *A*) may match
646 or exceed the original on this testbed; we leave it untested, because post-hoc
647 recomposition after seeing the map would not carry the evidential status of
648 the pre-specified chain.

649 **Why might gating be composition-bound?** We refrain from a mech-
650 anistic claim beyond the data, but note three compatible observations. First,
651 the gate-related differences concentrate on the minority, lowest-recall class
652 (Section 4.10), and matched-noise training lifts that class *only when the gate*
653 *is present*. One hypothesis, which we did not test, is that the gate is the
654 component through which matched-noise training improves hard-class recall,
655 i.e., that it exploits noise-aware training rather than acting as a test-time de-
656 noiser; we measured neither classification margins nor gate activations, and
657 the reading is in any case bounded by the matched-SNR scope (Section 6).
658 Second, the multiplicative form admits a per-sample suppression interpre-
659 tation, but no gate statistics were recorded, so this remains a hypothesis
660 rather than an observation. Third, the gateless composition falling to or
661 below S4D at the deepest noise levels is compatible with the other compo-
662 nents being calibrated to a gated regime (e.g., the stem may pass features
663 that are only safe when the gate can veto them); this is the weakest of the
664 three observations, since the gateless–S4D gap is not resolved with five seeds
665 (Section 4.9). All are testable in future work via pairwise interaction grids

666 and mismatched-noise protocols.

667 6. Boundaries and Limitations

668 We state the scope plainly.

- 669 1. **Testbed scope.** All claims are bounded to bearing vibration from
670 one pair of operating conditions of one dataset (37.5 Hz/11 kN and
671 40 Hz/10 kN; speed and load both change), binary classification, GPU
672 execution (no edge/real-time claims). The pair is run in both direc-
673 tions, but the reverse direction uses the same eight bearings with their
674 roles swapped and has a single IR bearing in its evaluation set, so it is
675 not an independent replication; the third condition of the dataset has
676 no IR bearing. The five seeds do not resample bearings (Section 3.1).
677 Independent bearings, multi-class settings and a second dataset are
678 future work.
- 679 2. **Attribution level.** Credit assigned to “the rest of the composition”
680 is by elimination within the enumerated component set; pairwise in-
681 teraction terms are not decomposed, and the host-by-gate contrast of
682 Section 4.4 is a four-arm estimate that cannot be assigned to a specific
683 component.
- 684 3. **The transplant conclusion is budget-conditional.** All arms share
685 a matched 50-epoch budget; the grafted gate trains normally yet re-
686 covers little, and convergence is not robustness (Section 4.4)—but we
687 cannot exclude that a longer or graft-tuned budget would recover more.
688 Our statement is therefore limited to *this gate branch, in this host, un-*
689 *der a matched training budget.*
- 690 4. **Matched-SNR scope.** Training and evaluation share the same in-
691 jected SNR throughout (Section 3.1); “robustness” in this paper is
692 matched-noise robustness under noise-aware training, not clean-train/noisy-
693 test generalisation. The gate’s benefit may operate by exploiting matched-
694 noise training (consistent with Section 4.10) rather than by test-time
695 denoising; attribution under mismatched-noise protocols is untested
696 and may differ.
- 697 5. **Graft dual reading.** Interaction dependence vs. transplant friction
698 (Section 4.4), the latter now subsumed under the budget condition of
699 item 3.

- 700 6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a
701 TCN baseline slightly overtakes the Mamba-style block, and -10 dB
702 is near-chance for the time-invariant baseline and excluded from ad-
703 judication; the demonstrated advantage regime is matched, Gaussian-
704 dominated noise.
- 705 7. **Conv-stem dual reading.** Its clean-data cost admits both representation-
706 mistransfer and optimisation-difficulty readings, reported in place (Sec-
707 tion 4.3).
- 708 8. **Reporting rule.** Reported values are final-epoch macro-F1 under a
709 fixed 50-epoch budget. The original grids’ best-epoch rule, for which
710 the thresholds of the early grids were pre-specified, is reported as a
711 sensitivity analysis and is optimistic by 0.6–10.6 pp (Section 4.9). Ev-
712 ery arm of the chain was re-run on the three adjudication levels; the
713 $+10/+6/clean/-10$ dB levels and the per-class grid were not re-run
714 and are auxiliary (best epoch). No source-domain validation rule was
715 available, and because the final epoch of the same evaluation condi-
716 tion has been inspected repeatedly, final-epoch values are not a fresh
717 confirmatory test.
- 718 9. **Scope of the freezing intervention.** The chain’s first step freezes
719 the input-dependent parts of Δ , A , B , C only; the fully input-independent
720 scan is a separate grid (Section 4.2) on the three adjudication levels,
721 which was specified after the earlier grids had been seen. No equivalence
722 margin was pre-specified for the cost of freezing, so “no measurable
723 cost” means the point estimates favour the frozen arms with intervals
724 including zero, not a demonstrated equivalence. Freezing is implement-
725 ed inside this Mamba-3 layer; a conclusion about input-dependence in
726 general would need other blocks and tasks.
- 727 10. **Gate interventions.** The gate-to-additive substitution has been made
728 in both hosts and both directions. In the Mamba-3 block the verdict
729 is *not rescued* on both transitions, with upper interval bounds of 0.486
730 and 0.362 against a threshold of 0.5 (the first close to it). In S4D
731 the width-matched arms are larger than the donor block, and on the
732 reverse transition the gate carries a small increment over the additive
733 branch that it does not carry on the original transition; this direction
734 dependence is unresolved.
- 735 11. **Strength of the pre-specification evidence.** Thresholds were pre-
736 specified internally; for all but the last three follow-up grids the sup-
737 porting evidence is local file times and run-directory names, not an

external time-stamped registry (Section 3.5). The process-level repeat check is a single observation.

12. **Construction checks.** For the frozen arm the state-dictionary check compares names and shapes of non-target tensors; value-level comparison was applied to the component-ablation and graft arms only.
13. **Second dataset.** A pre-specified replication on Paderborn real-damage bearings with disjoint training and evaluation bearings did not replicate the freeze or gate-removal contrasts, and the substitution test was not assessable there (Section 4.8); all arms transferred close to the majority-class level on that transition. The conclusions of this paper are therefore limited to XJTU-SY.
14. **Sample size.** Every cell mean uses five seeds. The smallest attainable one-sided Wilcoxon p is 0.031, paired intervals are wide, and several secondary statements (conv stem, A parameterisation, gateless versus S4D) are not resolved at this size; conclusions rest on effect sizes that clear the pre-specified thresholds, not on significance tests, which were never adjudication criteria.

A single-repeat check of process-level non-determinism (~ 0.7 pp) and the label-hash/adjudication trail are documented in the reproducibility appendices (Appendix A–Appendix B).

7. Conclusion

In this Mamba-3 block, on one controlled drift $\times$ noise testbed with matched-noise training, freezing the input-dependent Δ , A , B , C projections costs at most about 4 pp (an estimate that five seeds leave imprecise) while 15–20 pp of advantage over a time-invariant SSM remains, and freezing the trapezoidal weight and rotation angles as well—leaving no input-dependent tensor in the scan—costs nothing measurable and leaves 21–24 pp of advantage with fewer parameters. The multiplicative output gate is the one ablated component whose removal causes a large, one-signed loss (9–23 pp, returning the block to roughly the level of plain S4D), yet adding the gate to S4D recovers no more than the added structure does: 17–20% of the gap for a narrow graft, about 30% for one built in the donor’s width, and at least as much for an additive branch of the same size. Conversely, an additive branch of identical size and initial weights, placed where the gate was in the Mamba-3 block, recovers none of the loss on either transition. The

gate is thus necessary as a multiplicative operator in its native host, and what transfers to a time-invariant host is added structure rather than gating; the gate-related differences concentrate on the minority class. The first two steps replicate on the reverse transition. The evidence rests on one pair of operating conditions run in both directions with the same eight bearings, five seeds, and final-epoch reporting under a fixed budget, with the earlier evaluation-condition epoch selection retained as a sensitivity analysis. We offer the three-step chain—freeze, remove, add—as a protocol for turning architecture comparisons into component-level tests, with the caveat that the interventions must be matched in width and capacity for the add step to be conclusive, and we release our adjudication trail, including the incidents in which automated adjudication departed from the pre-specified criteria.

CRediT authorship contribution statement

Yi Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used LLM-based agents to execute and cross-check pre-specified analysis and adjudication passes as described in Appendix C, and to assist with manuscript drafting and figure generation. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. All numeric thresholds, adjudication rulings, and scientific interpretations reported in this paper were fixed and made by the human authors, as detailed in Appendix C.

803 Data availability

804 The code, pre-specification artifacts, and adjudication trail for this study
805 are available at <https://github.com/jeff-1980/bm3-jizi>. The repository
806 includes all follow-up grids (315 cells with per-epoch curves), their pre-
807 specifications, drivers, construction checks and decision memos and the har-
808 ness’s own imported modules, so the training entry point runs from a fresh
809 checkout.

810 Appendix A. Fairness protocol

811 *Appendix A.1. Evaluation-set fingerprint*

812 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
813 256 hash of the raw held-out test-label byte array, computed once per run
814 and asserted identical across every arm/condition/seed cell within a grid.
815 It fingerprints *that the same held-out label vector was scored every time*, not
816 model weights or predictions. It is also the split on which the best-epoch rule
817 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
818 five grids is:

819 `6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de`

820 This value is recomputed independently at adjudication time (not merely
821 copied from the run log) and cross-checked row-by-row against every grid’s
822 fairness report before any comparison across arms is treated as valid.

823 *Appendix A.2. Per-arm parameter counts*

824 Table A.12 lists every arm’s parameter count, each obtained directly via
825 `sum(p.numel() for p in model.parameters())` on the instantiated model
826 (not estimated), cross-verified against an independently maintained archi-
827 tecture map.

828 *Appendix A.3. Five-batch pre-specification timeline*

829 Table A.13 lists, for each of the five pre-specified grids, the artifact that
830 fixed its adjudication thresholds, that artifact’s creation/modification date,
831 and the corresponding grid’s run-directory start time. In every case the
832 prereg artifact predates the grid it governs.

833 Two honest caveats. First, the baseline batch predates the orchestration
834 infrastructure used for the other four batches by five days; its prereg trail

Table A.12: Per-arm parameter counts (verified against `arch_map.md` and harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802

Table A.13: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (mtime 18:26:54)	2026-07-02	2026-07-02 18:27
Component ablation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

835 rests on the task card’s creation timestamp alone, not on the same automat-
836 ed executor/reviewer machinery documented in [Appendix C](#), and we state
837 this rather than implying a uniform process across all five batches. Second,
838 for the component-ablation grid, two additional JSON files bearing “pre-
839 reg” in their name exist on disk (`secondary_prereg_20260704-0708` and
840 `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start
841 time; both are explicitly marked void in the corresponding decision artifact’s
842 `void_preregs` field, and neither was used to adjudicate any result in this pa-
843 per. They are the on-disk residue of the first post-hoc criterion substitution
844 described in [Appendix C](#), retained here (rather than deleted) as part of the
845 released adjudication trail.

846 **Appendix B. Confusion matrices and a repeat-run check**

847 *Appendix B.1. Full confusion-matrix set*

848 Table [B.14](#) gives the complete confusion-matrix set underlying the per-
849 class analysis of [Section 4.10](#): 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB),
850 each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These
851 counts were dumped per-sample and re-verified against the reported recal-
852 l/precision/F1 metrics cell-by-cell before use.

Table B.14: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true→predicted counts.

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	-6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
	-6 dB	15,634	1,886	2,543	6,177
S4D+gate	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
	-6 dB	13,267	4,253	3,244	5,476
frozen+gate	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
	-6 dB	13,200	4,320	5,469	3,251

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
S4D	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

853 *Appendix B.2. Repeat-run check*

854 Because two independent process runs of an identical configuration are
855 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
856 ductions in the selective-scan and normalisation ops), we measured a process-
857 level noise floor directly: the same arm, condition, and seed (`bm3_frozen`,
858 clean, seed 0, 2 epochs) was run twice in separate processes with identical
859 evaluation fingerprints and identical parameter counts confirmed. The two
860 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

861 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
862 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
863 run, and therefore neither an estimate nor a bound of process variability for
864 full-length runs. It is an order of magnitude below the seed-level variabil-
865 ity observed within the deep-noise band across the curve tables underlying
866 Figures 2 and 4 (standard deviations typically 2–9 pp, e.g. BM3-frozen 5.7–
867 9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Section 4
868 are judged against seed-level variability, not against this single repeat.

869 **Appendix C. Automation disclosure and adjudication gates**

870 *Appendix C.1. Automation disclosure*

871 The mechanical stages of the analysis pipeline—applying pre-specified nu-
872 meric thresholds to aggregated results, re-computing aggregates, re-verifying
873 evaluation fingerprints, and checking each decision against its pre-specified
874 rule—were carried out by LLM-based coding agents under an orchestration
875 harness, with every step auditable in the released trail. Manuscript drafting
876 and figure generation were also assisted by such agents (main text, Declara-
877 tion of Generative AI).

878 *Appendix C.2. Incidents*

879 Three incidents occurred in the original grids, each caught before an er-
880 roneous artefact entered a reported number: (1) on the frozen-selectivity
881 grid an automated decision substituted an unspecified criterion for the pre-
882 specified 10 pp threshold; re-applying the original rule gives the reported out-
883 come, “not the primary mechanism”; (2) on the component-ablation grid an
884 automated pass substituted an interval heuristic for the pre-specified thresh-
885 old, and post-hoc rule files were generated; the true pre-specification predated
886 the grid and was re-applied, and the post-hoc files are retained and marked
887 void; (3) on the graft grid a construction stage was falsely marked as passed;
888 the grid was re-executed from scratch under a corrected task card. Devia-
889 tions in the later follow-up grids (for example a bootstrap made deterministic
890 after first use, and rounding corrections) are recorded in the repository’s de-
891 cision memos. Full descriptions, with dates and artefact names, are given in
892 Supplementary Material S1.

893 *Appendix C.3. Human adjudication gates*

894 All numeric adjudication thresholds, every ruling on an incident, the de-
895 cision to report the incidents, and every interpretive claim in Section 5 were
896 fixed or made by the human authors. No agent could overwrite a protected
897 decision artefact, alter a pre-specified threshold or mark its own output as
898 final.

899 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

900 In line with Elsevier’s policy, AI tools are not listed as authors and the
901 human authors take full responsibility for the content. We extend the dis-
902 closure beyond the writing process to the adjudication pipeline because that
903 is where automation could affect the reported conclusions.

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